

Austrian zoologist Konrad Lorenz was a pioneer in the field of ethology (the study of animal behavior in natural environments). He is best known for his study of young birds, many of which, he discovered, behave in ways that are both innate (instinctive) and learned from the environment.

Born in Vienna, Austria, Konrad Lorenz was the son of a wealthy surgeon. At his father's wish, he studied medicine, but after qualifying, he went on to gain a PhD in zoology. Animals, especially birds, had always been his passion, and he built up a large menagerie as a child and a young man.

Innate vs. learned behavior

Before Lorenz, biologists made a clear distinction between two types of animal behavior: innate (instinctive) behavior, which is partly genetic, and learned behavior, which is dependent on experience. However, through his study of young birds, Lorenz showed that some behaviors are more complex. He highlighted a type of instinctive behavior that occurs at a particular age and is triggered by the environment.

Birds that can walk as soon as they hatch, such as ducks and geese, form bonds with the first moving object they see within a critical period. Usually this is their mother, but Lorenz discovered that newly hatched greylag geese could be tricked into bonding with him—and even a model train—in a process he later called “imprinting.” He observed that the courtship behavior of birds is similarly stage-linked and instinctive, following what he called “fixed-action patterns.” Since then, scientists have looked for similar patterns in the behavior of other animals, including humans.

Lorenz conducted his most famous behavioral experiments with greylag geese. The goslings that saw him before any other moving object when they hatched would instinctively bond with him as their “mother,” thereafter following him everywhere.

MILESTONES

IMPRINTING TESTS

Carries out his first experiment on geese in 1935, with more to follow on geese and jackdaws.

WARTIME DOCTOR

Serves as a doctor in the German army, is captured by the Soviets in 1944, and is imprisoned for 4 years.

POPULAR SCIENCE

Publishes *King Solomon's Ring* in 1949. Written for a popular audience, it is one of his most famous books.

NOBEL AWARD

Wins the 1973 Nobel Prize in Physiology or Medicine, jointly with Karl von Frisch and Nikolaas Tinbergen.

